

8 Robust Univariate Mean Estimation

8.1 Means and Medians

We begin with the simplest estimation problem: estimating the mean $\mu = \mathbb{E}_{X \sim \mathcal{D}}[X]$ of a distribution $\mathcal{D}$ over $\mathbb{R}$. We measure the quality of an estimate $\hat{\mu}$ using squared-error loss

$$\ell(\hat{\mu}; \mathcal{D}) = (\hat{\mu} - \mu)^2.$$

Given a clean sample $X = (X_1, \dots, X_n) \sim \mathcal{D}^n$, the most natural estimator is the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

The sample mean is not robust to adversarial corruption. Inserting one point can change the sample mean by an arbitrarily large amount.

The sample median is the classical robust alternative to the mean.

$$\text{med}(X) = \begin{cases} X_{(n/2)} & \text{if } n \text{ is even,} \\ X_{((n+1)/2)} & \text{if } n \text{ is odd,} \end{cases}$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics of the sample.

It is much less sensitive to extreme values, but without assumptions on the clean distribution it is not uniformly robust either. For example, the median of the sample $(0, 0, B)$ is 0, but after replacing 0 by B , the median is now B .

In fact, without any restriction on $\mathcal{D}$, robust mean estimation is impossible for every estimator. To see the basic obstruction consider

$$\mathcal{D}_0 = \delta_0 \quad \text{and} \quad \mathcal{D}_1 = (1 - \tau)\delta_0 + \tau\delta_B.$$

These distributions have means 0 and τB , which can be arbitrarily far apart. Nevertheless, an adversary in the mixture corruption model can make them identical by corrupting $\mathcal{D}_1$ to $\mathcal{D}_0$.

Intuitively, robust estimators try to identify and remove the corrupted data, at least in cases where that data is harmful. But there is no way to identify the corrupted points without some assumptions about what the uncorrupted points should look like. We will start with the natural starting point that the distribution is Gaussian: $\mathcal{D} = \mathcal{N}(\mu, \sigma^2)$. In this setting the sample median is robust because a Gaussian places a substantial amount of probability mass near its median, which is also its mean. An adversary must therefore corrupt a substantial fraction of the sample to move the median a substantial distance.

Theorem 8.1. *Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d. samples, and let $\tilde{X}$ be a τ -corruption of X . Then with probability at least $1 - \delta$ over the choice of the clean sample X ,*

$$(\mu_{\text{med}}(\tilde{X}) - \mu)^2 = O\left(\frac{\sigma^2 \log(1/\delta)}{n} + \sigma^2 \tau^2\right)$$

Proof sketch. Intuitively, the sorted clean sample lies near the $1/n, 2/n, \dots, n/n$ quantiles of the distribution. Corrupting at most τn points can move the median by at most τn positions in this sorted list, so the median of the corrupted sample lies between roughly the $(1/2 - \tau)$ and $(1/2 + \tau)$ quantiles of the clean distribution. Finally, if Φ is the standard Gaussian cumulative distribution function, then $\Phi^{-1}(1/2 + u) = O(u)$ for u bounded away from $1/2$. Thus the Gaussian quantiles at probability $1/2 \pm u$ are $\mu \pm O(\sigma u)$, and the total quantile error translates into

$$\left| \mu_{\text{med}}(\tilde{X}) - \mu \right| = O\left(\sigma \left(\tau + \sqrt{\frac{\log(1/\delta)}{n}} \right)\right).$$

Squaring the bound gives the result. The other term that is not $\sigma^2 \tau^2$ accounts for the noise due to random sampling. $\square$

The trimmed mean. The trimmed mean more explicitly implements the intuition of filtering outliers (with the caveat that “filtering outliers” isn’t really what is going on). Fix a constant $\gamma \in (0, 1/2)$, sort the corrupted sample as $\tilde{X}_{(1)} \leq \dots \leq \tilde{X}_{(n)}$, discard the largest and smallest $\lfloor \gamma n \rfloor$ points, and average the rest:

$$\mu_{\text{trim}}(\tilde{X}) = \frac{1}{n - 2\lfloor \gamma n \rfloor} \sum_{i=\lfloor \gamma n \rfloor + 1}^{n - \lfloor \gamma n \rfloor} \tilde{X}_{(i)}.$$

For Gaussian data and a corruption rate τ sufficiently smaller than γ , the remaining points lie within $O(\sigma)$ of μ , so the corrupted points that survive trimming can shift the estimate by only $O(\sigma\tau)$. Accounting for sampling error gives the same qualitative guarantee as the median:

$$\left(\mu_{\text{trim}}(\tilde{X}) - \mu \right)^2 = O\left(\frac{\sigma^2 \log(1/\delta)}{n} + \sigma^2 \tau^2 \right)$$

with probability at least $1 - \delta$.

The same estimator remains useful under a much weaker assumption, although the dependence on the corruption rate becomes worse.

Theorem 8.2 (Trimmed mean under bounded variance). *Suppose that $\mathcal{D}$ has mean μ and variance at most σ^2 , and that $\tilde{X}$ is a τ -corruption of n i.i.d. samples from $\mathcal{D}$. There is a choice of trimming level $\gamma = O(\tau + \log(1/\delta)/n)$ for which, provided $\gamma < 1/2$, with probability at least $1 - \delta$,*

$$\left(\mu_{\text{trim}}(\tilde{X}) - \mu \right)^2 = O\left(\sigma^2 \tau + \frac{\sigma^2 \log(1/\delta)}{n} \right).$$

Note that the term that depends on the corruption parameter is now $O(\sigma^2 \tau)$ instead of the $O(\sigma^2 \tau^2)$ that appears for Gaussian distributions.

Proof sketch. Set $\varepsilon = \tau + \log(1/\delta)/n$ and trim a sufficiently large constant multiple of εn observations from each end. Chebyshev’s inequality shows that a clean observation lies outside $[\mu - O(\sigma/\sqrt{\varepsilon}), \mu + O(\sigma/\sqrt{\varepsilon})]$ with probability $O(\varepsilon)$. A concentration bound for the number of such observations then shows that, with high probability, trimming removes every point outside this interval, including any

arbitrarily large corrupted points. Every corrupted point that remains has magnitude $O(\sigma/\sqrt{\epsilon})$ relative to μ , so their contribution to the average is at most $O(\sigma\tau/\sqrt{\epsilon}) = O(\sigma\sqrt{\epsilon})$. The clean observations removed by trimming have total bias at most $O(\sigma\sqrt{\epsilon})$ by Cauchy–Schwarz, and concentration of the remaining truncated observations contributes $O(\sigma\sqrt{\log(1/\delta)/n})$. Combining these bounds proves the claim. $\square$

Optimal robust rates. The different dependences on τ above are not artifacts of the analyses: they are the best possible, up to constant factors.

Theorem 8.3 (Optimal dependence on the corruption rate). *In one-dimensional mean estimation with known scale σ , the minimax robust squared error under τ -corruption is $\Theta(\sigma^2\tau^2)$ for Gaussian distributions and $\Theta(\sigma^2\tau)$ for the class of all distributions with variance at most σ^2 .*

Proof sketch. The upper bounds are given by the median (or the constant-fraction trimmed mean) for Gaussians and by the preceding trimmed-mean theorem for bounded variance distributions. For the matching Gaussian lower bound, take two Gaussians with variance σ^2 whose means differ by $c\sigma\tau$. Their total variation distance is $O(\tau)$, so an adversary can corrupt either distribution into the same observed distribution. No estimator can then determine which of the two means generated the data.

For bounded variance, compare the point mass at zero with

$$(1 - \tau)\delta_0 + \tau\delta_{\sigma/\sqrt{\tau}}.$$

Their total variation distance is τ , while their means differ by $\sigma\sqrt{\tau}$ and the second distribution has variance at most σ^2 . Again their corruption neighborhoods overlap, so the observed data cannot identify the clean distribution. Consequently every estimator has squared error $\Omega(\sigma^2\tau)$ on one of these two instances. $\square$

9 Robust Multivariate Mean Estimation

Before considering corruptions, let's recall what is possible with clean i.i.d. samples $X_1, \dots, X_n \sim \mathcal{D}$ in $\mathbb{R}^d$. Write $\mu = \mathbb{E}[X_i]$ and assume isotropic covariance $\Sigma = I_d$. Here I_d denote the $d \times d$ identity matrix. The isotropic assumption is the same as saying that each coordinate of X_i is sampled independently from $\mathcal{N}(0, 1)$. For an isotropic Gaussian $\mathcal{D} = \mathcal{N}(\mu, I_d)$, the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ satisfies

$$\bar{X} \sim \mathcal{N}(\mu, \frac{1}{n}I_d), \quad \mathbb{E}[\|\bar{X} - \mu\|_2^2] = \frac{d}{n}.$$

Gaussian concentration also gives, with probability at least $1 - \delta$,

$$\|\bar{X} - \mu\|_2^2 = O\left(\frac{d + \log(1/\delta)}{n}\right).$$

These bounds on the error are both optimal in the sense that no estimator can improve either bound on the error, so the sample mean is the best we can do for clean data.

If we assume only $\Sigma = I_d$, with no assumptions on higher moments, the expected squared error of the sample mean is still exactly d/n . The empirical mean will no longer give the high-probability bound that we get for Gaussian distributions, although there are other estimators that achieve the same high-probability statement. We now ask how these estimation guarantees change when an adversary can corrupt a fraction of the observations.

9.1 Resilience as a Paradigm for Robustness

Definition 9.1 (Resilience [SCV18]). A sample $S = (x_1, \dots, x_m)$ is (ρ, η) -resilient around μ if every subset $T \subseteq [m]$ with $|T| \geq (1 - \eta)m$ satisfies

$$\|\bar{x}_T - \mu\| \leq \rho, \quad \bar{x}_T = \frac{1}{|T|} \sum_{i \in T} x_i.$$

Here $\rho \geq 0$ and $0 \leq \eta < 1$. The center μ can be the sample mean or the population mean. Note that the norm $\|\cdot\|$ can be any norm we want in this definition.

Resilience obviously gives robustness to deletions: after deleting at most an η fraction of the sample, the mean of the remaining points is still within ρ of μ . More surprisingly, we can also handle insertions, although the algorithm we use becomes much more computationally expensive.

Theorem 9.2 (Robustness to arbitrary insertions [SCV18]). Let $\alpha \geq 0$ and $0 \leq \tau < 1/2$, and set

$$\eta = \frac{\tau}{1 - \tau}.$$

Suppose a clean sample S of $(1 - \tau)n$ points is $(\alpha/2, \eta)$ -resilient around μ in the norm $\|\cdot\|$. Let $\tilde{S}$ be obtained by inserting τn arbitrary points into S . There is an estimator using only $\tilde{S}$ and the parameters α, τ that outputs $\hat{\mu}$ satisfying

$$\|\hat{\mu} - \mu\| \leq \alpha$$

for every choice of the inserted points.

Proof sketch. Search for any subset $S' \subseteq \tilde{S}$ of size $(1 - \tau)n$ that is $(\alpha/2, \eta)$ -resilient around some center $\hat{\mu}$, and output $\hat{\mu}$. Such a subset exists because S itself qualifies. All subsets here refer to sample indices, so repeated values are allowed.

The intersection $T = S \cap S'$ satisfies

$$|T| \geq |S| + |S'| - n = (1 - 2\tau)n = (1 - \eta)|S| = (1 - \eta)|S'|.$$

Thus T is large enough for resilience of both sets to apply. Its mean is within $\alpha/2$ of both centers, so

$$\|\hat{\mu} - \mu\| \leq \|\hat{\mu} - \bar{x}_T\| + \|\bar{x}_T - \mu\| \leq \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha. \quad \square$$

Gaussian tails give particularly strong resilience in the Euclidean norm. The following statement includes the sampling error relative to the population mean.

Lemma 9.3 (Gaussian resilience in the Euclidean norm). *Let $X_1, \dots, X_m$ be i.i.d. samples from $\mathcal{N}(\mu, I_d)$, and fix $0 < \eta \leq 1/2$ and $\delta \in (0, 1/2)$. With probability at least $1 - \delta$, the sample $S = (X_1, \dots, X_m)$ is (ρ, η) -resilient around μ in $\|\cdot\|_2$, where*

$$\rho = O\left(\eta\sqrt{\log(1/\eta)} + \sqrt{\frac{d + \log(1/\delta)}{m}}\right).$$

Equivalently, every subset $T \subseteq [m]$ with $|T| \geq (1 - \eta)m$ simultaneously satisfies $\|\bar{X}_T - \mu\|_2 \leq \rho$.

Combining this with Theorem 9.2, using $m = (1 - \tau)n$ and $\eta = \tau/(1 - \tau)$, gives an estimator under arbitrary insertions with Euclidean error

$$\|\hat{\mu} - \mu\|_2 = O\left(\tau\sqrt{\log(1/\tau)} + \sqrt{\frac{d + \log(1/\delta)}{n}}\right)$$

with probability at least $1 - \delta$.

9.2 The Tukey Median

In one dimension, a median has at least half the observations on either side. In higher dimensions, we can ask how well a point balances the sample in every direction. This leads to Tukey depth, also called *halfspace depth*.

Definition 9.4 (Tukey depth). For a sample $X = (X_1, \dots, X_n)$ in $\mathbb{R}^d$, the Tukey depth of a point $z \in \mathbb{R}^d$ is

$$\text{depth}(z; X) = \inf_{v \in \mathbb{R}^d: \|v\|_2=1} \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\langle v, X_i - z \rangle \geq 0\}.$$

Equivalently, it is the smallest fraction of the sample in any closed halfspace containing z .

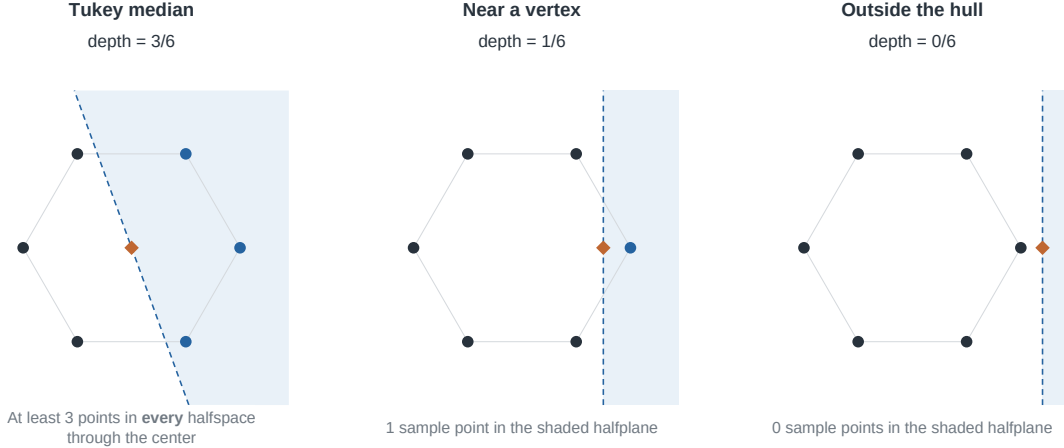


Figure 4: Tukey depth for six points at the vertices of a regular hexagon. Orange diamonds mark the query points; each shaded halfspace attains the minimum count for its query point.

For a distribution $\mathcal{D}$, define Tukey depth by

$$\text{depth}(z; \mathcal{D}) = \inf_{v \in \mathbb{R}^d: \|v\|_2=1} \mathbb{P} [\langle v, Y - z \rangle \geq 0].$$

A *Tukey median* is any point of maximum depth:

$$\mu_{\text{Tukey}}(X) \in \arg \max_{z \in \mathbb{R}^d} \text{depth}(z; X).$$

It need not be a sample point or be unique. In one dimension this recovers the usual median, including the interval of medians between the two middle observations when n is even.

Depth for Gaussians. For $\mathcal{D} = \mathcal{N}(\mu, I_d)$, every projection onto a unit vector is a one-dimensional Gaussian with variance 1. The halfspace with the least mass points away from μ , giving

$$\text{depth}(z; \mathcal{D}) = \Phi(-\|z - \mu\|_2),$$

where Φ is the standard Gaussian cumulative distribution function. Thus μ has depth 1/2, and points far from μ have small depth.

Theorem 9.5 (Robustness of the Tukey median for Gaussians). *Let $X_1, \dots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, I_d)$. There is a universal constant $c > 0$ such that, if*

$$\tau + \sqrt{\frac{d + \log(1/\delta)}{n}} \leq c,$$

then with probability at least $1 - \delta$ over the clean sample, simultaneously for every sample $\tilde{X}$ obtained by replacing at most τn of the dataset, and every Tukey median of $\tilde{X}$,

$$\left\| \mu_{\text{Tukey}}(\tilde{X}) - \mu \right\|_2^2 = O\left(\tau^2 + \frac{d + \log(1/\delta)}{n} \right).$$

Proof sketch. The family of halfspaces in $\mathbb{R}^d$ has VC dimension $d + 1$. Uniform convergence therefore bounds the difference between empirical and population halfspace probabilities by

$$\beta = O\left(\sqrt{\frac{d + \log(1/\delta)}{n}}\right)$$

with probability at least $1 - \delta$. Replacing τn points changes the fraction in any halfspace by at most τ . Hence, uniformly in z ,

$$\left| \text{depth}(z; \widetilde{X}) - \text{depth}(z; \mathcal{D}) \right| \leq \beta + \tau.$$

Write $\widehat{\mu} = \mu_{\text{Tukey}}(\widetilde{X})$. Since $\widehat{\mu}$ maximizes empirical depth and the population depth of μ is $1/2$,

$$\text{depth}(\widehat{\mu}; \mathcal{D}) \geq \text{depth}(\widehat{\mu}; \widetilde{X}) - (\beta + \tau) \geq \text{depth}(\mu; \widetilde{X}) - (\beta + \tau) \geq \frac{1}{2} - 2(\beta + \tau).$$

The Gaussian depth formula then implies

$$\|\widehat{\mu} - \mu\|_2 \leq \Phi^{-1}\left(\frac{1}{2} + 2(\beta + \tau)\right) = O(\beta + \tau)$$

when $\beta + \tau$ is sufficiently small. □

The corruption-dependent error has no dependence on d ; dimension enters through sampling error. Computing a Tukey median in high dimensions, however, is computationally expensive, motivating the efficient algorithms we discuss next.